



2IA Recap Problems

It is that time of the year ...



Recap problems

1. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**
 - (a) $\mathbb{Z}$ and $\mathbb{Q}$ are isomorphic.
 - (b) The cyclic group $\mathbb{Z}_n$ has exactly n generators.
 - (c) The intersection of any two subgroups of a group G can never be the empty set.
 - (d) $\mathbb{Z}_2$ is a subgroup of $\mathbb{Z}_4$.
 - (e) Every two cyclic groups of the same order are isomorphic.
 - (f) There exists group G that is isomorphic to a proper subgroup of itself.

Solution:

(a) **False:** If $\mathbb{Q}$ were isomorphic to $\mathbb{Z}$, then $\mathbb{Q}$ would be cyclic; that is, $\mathbb{Q} = \langle q \rangle = \{kq \mid k \in \mathbb{Z}\}$ for some $q \in \mathbb{Q}$. But then $q^2 = nq$ for some $n \in \mathbb{Z}$. Using the cancellation laws in $\mathbb{Q}$, we would get that $q = n \in \mathbb{Z}$; which implies:

$$\mathbb{Q} = \langle q \rangle = \{kq \mid k \in \mathbb{Z}\} \subseteq \mathbb{Z},$$

a contradiction.

(b) **False:** Consider, for example, $\mathbb{Z}_4 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$. Recall that $\bar{a}$ is a generator of $\mathbb{Z}_4$ if and only if $\gcd(a, 4) = 1$. From here, we obtain that $\mathbb{Z}_4$ has two generators, namely, $\bar{1}$ and $\bar{3}$.

(c) **True:** Every subgroup contains the identity element of G , so the intersection of any two subgroups contains at least the identity element.

(d) **False:** A subgroup (is by definition) a subset of a group such that it is a group itself with the same operation. In this case, $\mathbb{Z}_2$ is **NOT** a subset of $\mathbb{Z}_4$, and so it does not make sense to think about subgroups.

Be very careful here, we are using the same notation here to denote $\bar{0} \in \mathbb{Z}_2$ and $\bar{0} \in \mathbb{Z}_4$ but these are not the same! $\bar{0}$ viewed as an element of $\mathbb{Z}_2$ denotes the even numbers, while $\bar{0} \in \mathbb{Z}_4$ denotes all the integers that have a remainder of 0 when divided by 4; these two sets are not equal!

(e) **True:** This follows from the Isomorphism Theorem of Cyclic Groups: every cyclic group of order n is isomorphic to $\mathbb{Z}_n$, while every cyclic group of order infinite is isomorphic to $\mathbb{Z}$.

(f) **True:** Take $G = \mathbb{Z}$ and $H = 2\mathbb{Z}$. We have that H is a cyclic group itself because every subgroup of a cyclic group is cyclic. Now since H is an infinite cyclic group, it must be isomorphic to $\mathbb{Z}$, by the Isomorphism Theorem of Cyclic Groups.

2. Let G be a group. Prove that G is abelian if and only if $(ab)^2 = a^2b^2$ for all $a, b \in G$.

Solution: Suppose first that G is abelian. Then, for any $a, b \in G$, using that the operation in G is associative and that $ab = ba$, we have that

$$(ab)^2 = (ab)(ab) = a(ba)b = a(ab)b = a^2b^2,$$

as desired. Conversely, assume now that $(ab)^2 = a^2b^2$ for all $a, b \in G$. Take any $a, b \in G$. Then, using that a and b both have inverses in G we obtain that

$$\begin{aligned} abab &= (ab)^2 = a^2b^2 \Rightarrow a^{-1}(abab) = a^{-1}(a^2b^2) \Rightarrow (a^{-1}a)(bab) = (a^{-1}a^2)b^2 \\ &\Rightarrow bab = ab^2 \Rightarrow (bab)b^{-1} = (ab^2)b^{-1} \Rightarrow (ba)(bb^{-1}) = a(b^2b^{-1}) \\ &\Rightarrow ba = ab, \end{aligned}$$

proving that a and b commute. Since this holds for any two elements in G , we can conclude that G is abelian.

3. Let G be a group and $z \in Z(G)$. Write G^z to denote the set G with the operation $a * b = abz^{-1}$. Prove the following assertions:

- (a) G^z is a group.
- (b) $G \cong G^z$.

Solution: (a) First all of notice that $*$ satisfies the associativity law since G is a group (so the operation that we have in G is associative). The identity element in G^z is z ; in fact, for $g \in G^z$ we have that

$$g * z = gzz^{-1} = g, \quad z * g = zgz^{-1} \stackrel{z \in Z(G)}{=} gzz^{-1} = g.$$

Lastly, the inverse of g in G^z is $g^{-1}z^2$. In fact,

$$\begin{aligned} g * g^{-1}z^2 &= g(g^{-1}z^2)z^{-1} = (gg^{-1})(z^2z^{-1}) = z, \\ g^{-1}z^2 * g &= g^{-1}z^2gz^{-1} \stackrel{z \in Z(G)}{=} z^2(g^{-1}g)z^{-1} = z^2z^{-1} = z. \end{aligned}$$

(b) Define $\varphi : G \rightarrow G^z$ by $\varphi(g) = gz$. Notice that φ is bijective since its inverse is $\psi : G^z \rightarrow G$ given by $\psi(g) = gz^{-1}$ (please, complete the details; that is, check that $\varphi \circ \psi = \text{Id} = \psi \circ \varphi$). It remains to check that it is a group homomorphism; to do so, take $a, b \in G$

$$\begin{aligned} \varphi(a) * \varphi(b) &= az * bz = (az)(bz)z^{-1} = abz, \\ \varphi(ab) &= (ab)z \stackrel{z \in Z(G)}{=} abz. \end{aligned}$$

4. Let G be a group with order $|G| < 400$. Suppose that G has subgroups of orders 45 and 75 and find $|G|$.

Solution: By Lagrange's Theorem, we get that $45 \mid |G|$ and $75 \mid |G|$. This tells us that $|G|$ is a common multiple of 45 and 75. Notice that $\text{lcm}(45, 75) = 3^2 \cdot 5^2 = 225$. Since the order of G must be a multiple of 225 and it has to be less than 400, we can conclude that $|G| = 225$.

5. Let H and K be subgroups of a group G such that $|H| = p$, where p is a prime number. Prove that either $H \subseteq K$ or $H \cap K = \{1\}$.

Solution: Notice that H is a finite group, so we can apply Lagrange's Theorem to $H \cap K$ to obtain that $|H \cap K|$ divides $|H|$. But since we are assuming that $|H| = p$ is prime, we get that either $|H \cap K| = 1$ or $|H \cap K| = p = |H|$. If $|H \cap K| = 1$, then $H \cap K = \{1\}$; if $|H \cap K| = p = |H|$, then $H = H \cap K \subseteq K$, finishing the proof.

6. Let G be a group, H, K subgroups of G and $a, b \in G$. Prove that either $Ha \cap Kb = \emptyset$ or $Ha \cap Kb = (H \cap K)c$ for some $c \in G$.

Solution: Suppose that $Ha \cap Kb \neq \emptyset$ and let us show that $Ha \cap Kb = (H \cap K)c$ for some $c \in G$. Since $Ha \cap Kb \neq \emptyset$, there exists $c \in Ha \cap Kb$. Now, using what we proved in the workshops, we know that $(H \cap K)c = Hc \cap Kc$.

On the other hand, since $c \in Ha$, using the coset properties we obtain that $Hc = Ha$; similarly, using now that $c \in Kb$ we get that $Kc = Kb$. Putting altogether we have that

$$(H \cap K)c = Hc \cap Kc = Ha \cap Kb,$$

as required.

7. Let K be a subgroup of G of index 2.
- (a) Determine the right cosets of K in G .
 - (b) Prove that K is normal in G
 - (c) If $a \in G \setminus K$ and $b \in G \setminus K$, prove that $ab \in K$.

Solution: (a) From $|G : K| = 2$, we get that there are two cosets of K in G . One of them is K , and since the cosets form a partition of G , the other one must be $G \setminus K$. Thus, the cosets are $\{K, G \setminus K\}$.

(b) For $g \in G$, we have to prove that $Kg = gK$. We consider two cases:

- Case 1: $g \in K$

In this case $Kg = K$ and $gK = K$. Thus, $Kg = gK$.

- Case 2: $g \notin K$. Then $g \in G \setminus K$ and so $Kg = G \setminus K = gK$.

In any case, we have proved that $Kg = gK$, and since this is for all $g \in G$, we have that K is normal in G .

(c) By (a) we know that the (right or left) cosets of K in G are $\{K, G \setminus K\}$. From $a \notin K$, we obtain that $Ka = G \setminus K$ and since $b^{-1} \notin K$, we get that $Kb^{-1} = G \setminus K$. This shows that

$$Ka = Kb^{-1},$$

which implies that $K(ab) = K$, and so $ab \in K$ by the cosets properties.

8. Let H be a subgroup of a group G such that if $a, b \in G$ satisfy $Ha \neq Hb$, then $aH \neq bH$. Prove that H is normal in G .

Solution: Suppose that H is not normal in G . Then by the Normality Test we can find $g \in G$ such that $gHg^{-1} \not\subseteq H$. Thus, there exists $h \in H$ such that $ghg^{-1} \notin H$. The cosets properties yields that $H(ghg^{-1}) \neq H$, which implies that $H(gh) \neq Hg$. But then $(gh)H \neq gH$ by the assumption on H . On the other hand using that $hH = H$, we get

$$gH \neq (gh)H = g(hH) = gH,$$

a contradiction.

9. Let G be a group and $\varphi : G \rightarrow G$ an automorphism of G ; that is, a group homomorphism that is bijective.
- (a) Prove that the subset $H = \{g \in G \mid \varphi(g) = g\}$ consisting on all the elements of G fixed by φ is a subgroup of G .
 - (b) Let G be finite now. Suppose that $\varphi : G \rightarrow G$ fixes more than half of the elements of G and show that φ is the identity automorphism; that is, $\varphi(g) = g$ for all $g \in G$.

Solution: (a) To prove that H is a subgroup of G , we use the Subgroup Test. Notice that $1 \in H$ because $\varphi(1) = 1$, since we are assuming that φ is a homomorphism. Now, for $a, b \in H$, using that φ is a group homomorphism and the definition of H we get that

$$\begin{aligned}\varphi(ab) &= \varphi(a)\varphi(b) = ab \Rightarrow ab \in H \\ \varphi(a^{-1}) &= \varphi(a)^{-1} = a^{-1} \Rightarrow a^{-1} \in H\end{aligned}$$

We can then conclude that H is a subgroup of G .

(ii) Assume now that G is finite, and that $\varphi : G \rightarrow G$ fixes more than half of the elements of G . From (a) we have that H is a subgroup of G that contains more than half of the elements of G , and since $|H|$ divides $|G|$ (by Lagrange's Theorem) we can conclude that $H = G$, as required.